Anton Kundenko

Principal Systems Engineer • Database Architect • Programming Language Creator

Designing ultra-fast analytical systems that operate at hardware speed — SIMD-first, cache-aware, and minimal by construction. Author of **Rayfall** (Lisp-like array language for vectorized analytics) and **O** (real-time streaming & historical data processing).

hetoku.com | github.com/singaraiona | LinkedIn

SUMMARY

Principal-level systems engineer and database architect with **18+ years** across embedded systems, mobile OS/ARM platform work, and financial technology. Founder and lead architect of **RayforceDB** (SIMD-optimized columnar database + **Rayfall** language ecosystem) and founder of **TeideDB** (analytical SQL engine / embedded analytics runtime). Co-founder of **ThePlatform**, a real-time analytics stack with custom language/runtime for streaming and historical data processing. Deep expertise in SIMD optimization (AVX2/NEON), cache-aware layouts, mmap-based storage, lock-free concurrency, and low-latency market data processing.

KEY PROJECTS

Teidelum — Founder & Lead Architect

2025 — Present · lum.teidedb.com

Open-source, single-binary knowledge infrastructure built on the Teide columnar engine. Syncs work tools, indexes content, and serves data to AI agents via MCP and applications via REST — entirely local, no cloud dependencies.

- BM25-ranked full-text search with fuzzy matching and highlighted snippets.
- SQL analytics powered by the embedded Teide columnar engine — filtering, grouping, sorting, and joins over structured data.
- Graph traversal via foreign-key relationships; dual API (MCP over stdio + REST over HTTP) exposing 11 tools.
- Incremental sync from Notion, Zulip, and live database connections; one Rust binary, no Docker or config files.

Rust · Columnar engine · MCP · Full-text search · SQL · MIT licensed

TeideDB — Founder & Lead Architect

2024 — Present · teidedb.com

Analytical SQL engine / embedded analytics runtime designed to bring a modern SQL interface to a vectorized, in-process execution model.

- Vectorized operators and optimizer-oriented architecture for fast analytical workloads.
- Designed for agent-driven analytics pipelines and interactive reporting over columnar data.
- Integrates with the broader Rayforce ecosystem (storage, runtime components, and tooling).

SQL · Vectorized execution · Embedded / in-process · C · Performance engineering

AXL DB — Founder

2021 — Present · axl-db.com

Ultra-compact (<1MB core) zero-dependency vector database engine optimized for deterministic latency.

- SIMD-first interpreter with cache-aware layouts and batched I/O.
- Designed for high-frequency trading, edge analytics, and embedded deployments across ARM/x86.

Vector DB · <1MB core · Deterministic latency · AVX2/NEON

RayforceDB — Founder & Lead Architect

2020 — Present · rayforcedb.com

SIMD-first columnar analytical database with a Lisp-like query language (**Rayfall**), implemented in C with zero dependencies.

- Designed a vectorized execution model with cache-friendly column layouts and mmap-based storage.
- Built an end-to-end ecosystem: core engine, language/runtime, test & benchmark tooling, and bindings (Python/Rust/WASM).
- Targeted at time-series analytics and low-latency decision pipelines as a modern alternative to kdb+ / embedded analytics stacks.

C17 · SIMD (AVX2/NEON) · Columnar · mmap · Zero deps · Language design

FOCUS

DOMAINS

Columnar analytics • Time-series • Embedded analytics • Low-latency finance

STRENGTHS

SIMD optimization • Cache-aware design • Language runtimes • Minimal/zero-dep systems

WORKING STYLE

Build-from-scratch architectures, correctness under load, clean minimal code, measurable performance.

TECHNICAL SKILLS

LANGUAGES

C (C17) · Rust · Python · k/q (kdb+) · Lisp/Scheme · Rayfall · O language

PERFORMANCE

SIMD (AVX2/NEON) · Profiling (perf) · Cache optimization · Zero-copy · Memory layouts

DATABASES

Columnar storage · Vectorized execution · Query runtimes · Index structures · Time-series

SYSTEMS

Linux · ARM · mmap I/O · Concurrency · Lock-free

TOOLING

Git · Make/CMake · Sanitizers · Benchmarks · CI

OPEN SOURCE

Maintains multiple performance-focused open-source projects (databases, runtimes, concurrency libraries).

GitHub: github.com/singaraiona

LANGUAGES

English (professional) • Russian (native) • Ukrainian (fluent)

ThePlatform — Co-Founder

2016 — Present · theplatform.technology

Real-time + historical analytics stack for financial data with a custom programming language (O) and async messaging runtime.

- Built language/runtime components and a declarative model for streaming & historical analytics.
- Implemented communication reagents: TCP/UDP/WebSocket/IPC/TLS, timers, plugins, and standard library tooling (parsers, serialization).
- Used to power low-latency analytics workflows and risk/market-data pipelines.

Language + runtime · Time-series · Low-latency · C · Streaming

hyperbridge

Rust · GitHub

High-throughput multi-producer/multi-consumer unbounded channel with async support (lock-free, concurrency-first).

Rust · Lock-free · Concurrency · Async

EXPERIENCE

Founder & Principal Architect

2020 — Present

RayforceDB / TeideDB

- Architected multiple database engines and language runtimes focused on vectorized execution and minimal footprint.
- Designed SIMD-optimized operators, memory layouts, storage formats, and execution models for analytical workloads.
- Built production-grade build & test infrastructure and public open-source ecosystems (bindings, benchmarks, docs).

Co-Founder

2016 — Present

ThePlatform

- Built a real-time analytics stack used in financial data processing environments.
- Designed language/runtime capabilities enabling declarative analytics and streaming computation.

Senior Engineer

2013 — 2020

Confidential (NDA) — Algorithmic Trading / FinTech

- Developed low-latency trading infrastructure, market-data processing, and real-time analytics pipelines.
- Worked on performance-critical systems where deterministic latency and correctness under load are essential.

Lead Engineer, Project Lead

2012 — 2013

Samsung Electronics Ukraine R&D

- Led OS virtualization work on ARM Cortex-A15; implemented TrustZone security extensions and DRM systems.
- Delivered low-level platform software integrating kernel, security, and hardware-software boundaries.

Embedded / Systems Developer

2007 — 2012

TranSat • Technotrade

- Built embedded systems (smart house, voting, CCTV, parking control) and POS/payment terminal software.
- Developed transaction-critical systems with strong reliability requirements.

EDUCATION

Specialist Degree (Master-level) — Automation & Computer-Integrated Technologies

Donbass State Engineering Academy • 2001 — 2006

Automation engineering foundation: embedded systems, computer-integrated technologies, and systems programming.